



8281 - A gravitational perturber in B1938+666: a challenge for CDM or an unusual globular cluster?

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1		NIRCam Imaging	(1) B1938+666

ABSTRACT

Cold Dark Matter (CDM) is the current standard paradigm for galaxy formation and successfully reproduces a wide variety of observations. Still, it remains largely untested in the non-linear sub-galactic regime. A clear prediction of this model is that many low-mass halos should surround any galaxy and should populate its line of sight. As most of these objects are expected to be completely dark, strong gravitational lensing provides a unique channel to detect them and determine their properties.

JWST Proposal 8281 (Created: Friday, June 13, 2025, 2:00:10PM Eastern Standard Time) - Overview

Very long baseline interferometric observations at 1.7 GHz of the gravitational lens system B1938+666 have led to the gravitational detection of an unprecedentedly low-mass object of unknown nature. With a total mass of just $10^6 M_{\text{sun}}$, it could be a non-luminous CDM halo, a globular cluster, or an ultra-compact dwarf galaxy. However, its inferred properties are inconsistent with those predicted for CDM dark matter halos, and unusual when compared to globular clusters in the Local Universe. Its high concentration agrees better with that of a core-collapsed dark matter halo, as is possible in self-interacting dark matter models.

The JWST observations requested here are the final and crucial ingredient in determining the nature of this detected object. Measuring or placing a meaningful lower limit on its mass-to-light ratio is the most reliable avenue to robustly distinguish between a globular cluster, a highly concentrated dark matter halo, and an ultra-compact dwarf galaxy. These JWST observations will unleash the full impact of this remarkable discovery, with implications either for the nature of dark-matter or for the evolution of compact stellar systems across cosmic time.

OBSERVING DESCRIPTION

We request observations with NIRCам in the F115W filter to image the strong gravitational lens system B1938+666 (lens redshift 0.881, source redshift 2.059). We will use these observations to measure the mass-to-light ratio of a $10^6 M_{\text{sun}}$ object, which has been recently gravitationally detected from very long baseline interferometric observations at 1.7 GHz. This measurement will allow us to test whether the detected object is a globular cluster, an ultra-compact dwarf galaxy, or a dark-matter halo. In the latter case the cold dark matter paradigm will be excluded.

Unlike existing Keck, HST/WFC2 and HST/NICMOS data of the same system, these JWST observations will provide the necessary depth, angular resolution and wavelength coverage to achieve our scientific goal. The choice of filter is motivated by the need to maximise the angular resolution and the contrast between the emission from the lensing galaxy and Einstein ring, and that from the detected object.

Under the assumption that the detected object is a globular cluster (see proposal for more details), we have estimated a conservative lower limit on its apparent AB magnitude of 30.15. The exposure time is calculated to reach a detection threshold of 10 sigma, once the shot noise from the Einstein ring and the lensing galaxy emission has been modelled and subtracted. From simulations, we find that this is reached in 24.6 hours of exposure time (30.4 hours in total).

We apply a standard dither pattern to mitigate detector defects and improve the final resolution of the images.

Proposal 8281 - Targets - A gravitational perturber in B1938+666: a challenge for CDM or an unusual globular cluster?

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	B1938+666	RA: 19 38 25.2800 (294.6053333d) Dec: +66 48 52.92 (66.81470d) Equinox: J2000		
	Comments: Category=Galaxy Description=[Einstein rings] Extended=YES				

Proposal 8281 - Observation 1 - A gravitational perturber in B1938+666: a challenge for CDM or an unusual globular cluster?

Observation	Proposal 8281, Observation 1										
	Diagnostic Status: Warning Observing Template: NIRCam Imaging										
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 1:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	B1938+666	RA: 19 38 25.2800 (294.6053333d) Dec: +66 48 52.92 (66.81470d) Equinox: J2000								
	Comments: Category=Galaxy Description=[Einstein rings] Extended=YES										
Template	Module		Subarray				Target Placement				
	ALL		FULL				Module Gap				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRAMODULE		3		STANDARD				7	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID	
	1	F115W	F277W	MEDIUM8	8	5	105	21	88836.035		
Special Requirements	Group Visits within 53.0 Days Visits Same PA Offset 120.42 arcsec, 35.35 arcsec Background Limited. Background no more than 10th percentile above minimum										